

BOX 2026 – Homework 1

September 2, 2026

Instructions

1. Download the genome file `hw1.fa` from:

<https://drive.google.com/drive/folders/1jna1SwlrnAKLHUGjrfL5Bhbt4EG90Nu>

2. Identify the species represented by the genome. Select a sufficiently long substring of the genome and search for it using NCBI BLAST <https://blast.ncbi.nlm.nih.gov/Blast.cgi>. Report the species you identify and briefly explain how you reached your conclusion.

3. In Python, compute the *subword complexity*

$$F(k) = \text{number of distinct substrings of length } k$$

for the genome, for every $k = 2, \dots, 30$. Plot $F(k)$ as a function of k .

4. Generate a random DNA sequence of the same length as the genome, using the alphabet $\{A, C, G, T\}$. Compute and plot its subword complexity for $k = 2, \dots, 30$.
5. Construct a prefix of the infinite Fibonacci word having the same length as the genome. The Fibonacci word is obtained by starting from A and repeatedly applying

$$A \rightarrow AB, \quad B \rightarrow A.$$

For example,

$$A, \quad AB, \quad ABA, \quad ABAAB, \quad ABAABABA, \dots$$

Compute and plot its subword complexity for $k = 2, \dots, 30$.

6. Compare the three subword-complexity functions:

- the genome,
- the random DNA sequence,
- the Fibonacci word.

Describe the main differences you observe and explain what they indicate about the structure of these three sequences.

Submission

Submit your code through a personal GitHub repository.

- Repository name: `box2026-yourSurname`
- Example: `box2026-brassens`
- Add `karel-brinda` and `leo-ackermannn` as collaborators.

Your repository should contain the Python code needed to reproduce the calculations and plots, together with a short README describing how to run it.